Companion XXV

The 21-cm Signature of the Relaxation-Driven Cyclic Cosmology Reionization Topology, X-ray Heating, and Large-Scale Structure

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Abstract

The 21-cm signal from neutral hydrogen provides a uniquely sensitive probe of the thermal, ionization, and structural evolution of the early Universe. Upcoming observations from SKA, HERA, and LOFAR will map the morphology of reionization, the timing of X-ray heating, and the large-scale distribution of ionized and neutral regions. These measurements offer a powerful test of the Relaxation-Driven Cyclic Cosmology (RDCC), whose modified small-scale structure, delayed fragmentation, and enhanced early black hole and quasar activity fundamentally alter the topology of reionization.

Building on the early-galaxy physics of Companion XXII, the black hole formation framework of Companion XXIII, and the quasar radiative feedback model of Companion XXIV, we derive the RDCC predictions for the global 21-cm signal, the 21-cm power spectrum, and the morphology of ionization bubbles. We show that RDCC produces smoother, more coherent ionization structures, delayed but stronger X-ray heating, and a characteristic suppression of small-scale 21-cm power due to the relaxation-modified halo mass function.

This Companion establishes the 21-cm signal as a direct observational probe of the relaxation scale $k_{\rm rel}$ and provides clear predictions for SKA, HERA, LOFAR, JWST, and cross-correlation studies with early galaxies and quasars.

1 Motivation and Overview

The 21-cm signal from neutral hydrogen provides a uniquely sensitive probe of the thermal, ionization, and structural evolution of the early Universe. Unlike galaxy surveys or quasar observations, which trace only the brightest objects, the 21-cm line maps the entire intergalactic medium (IGM), including faint galaxies, diffuse gas, and the topology of reionization. Upcoming observations from SKA, HERA, and LOFAR will measure the global 21-cm signal, the 21-cm power spectrum, and the morphology of ionized and neutral regions across cosmic time.

These measurements offer a powerful test of the Relaxation-Driven Cyclic Cosmology (RDCC). The suppression of small-scale power, delayed fragmentation, and enhanced early black hole and quasar activity predicted by RDCC fundamentally alter the timing, morphology, and coherence of reionization. As shown in **Companion XXII**, early star formation is delayed and smoother; in **Companion XXIII**, black hole seeds are more massive and grow more efficiently; and in **Companion XXIV**, early quasars produce larger and more coherent ionization bubbles.

The 21-cm signal is therefore a direct tracer of the relaxation scale $k_{\rm rel}$ and provides a unique opportunity to test the RDCC framework.

1.1 Observational Context

Current and upcoming experiments probe different aspects of the 21-cm signal:

- HERA measures the 21-cm power spectrum at $z \sim 6$–20,
- LOFAR constrains large-scale ionization structure at $z \sim 8$–10,
- SKA will map the full 3D morphology of reionization,
- JWST provides complementary constraints on early galaxies and quasars,
- cross-correlations (21-cm $\times$ galaxies $\times$ quasars) probe bubble topology.

These observations are sensitive to:

- the timing of X-ray heating,
- the growth of ionized bubbles,
- the clustering of early sources,
- the small-scale structure of the IGM,
- the global reionization history.

Each of these quantities is modified in RDCC.

1.2 RDCC Perspective

The Relaxation-Driven Cyclic Cosmology modifies early structure formation through the scale-dependent relaxation rate $\alpha(a)$, which suppresses small-scale power and delays the collapse of low-mass halos. As shown in **Companion XVIII** and **Companion XIX**, this leads to:

- reduced small-scale clumping,
- smoother density fields,
- delayed but more coherent star formation,
- enhanced early black hole and quasar activity,
- larger and smoother ionization bubbles.

These effects directly influence the 21-cm signal:

- **Global signal:** delayed absorption trough, smoother heating transition,
- **Power spectrum:** suppressed small-scale power, enhanced large-scale modes,
- **Bubble topology:** larger, more coherent ionized regions,
- **Cross-correlations:** stronger 21-cm $\times$ quasar anticorrelation.

Thus, RDCC predicts a distinctive 21-cm signature that differs qualitatively from ΛCDM.

1.3 Goals of This Companion

The goals of this Companion are:

- to derive the RDCC predictions for the global 21-cm signal,
- to compute the RDCC-modified 21-cm power spectrum,
- to model the morphology of ionization bubbles in RDCC,
- to analyze the impact of early quasars on X-ray heating and reionization,
- to compare RDCC predictions with SKA, HERA, LOFAR, and JWST.

These results build directly on the early-galaxy, black hole, and quasar physics developed in Companions XXII–XXIV.

1.4 Structure of This Companion

The structure of this Companion is as follows:

- Section 2 derives the RDCC global 21-cm signal,
- Section 3 computes the RDCC 21-cm power spectrum,
- Section 4 analyzes the morphology of ionization bubbles,
- Section 5 develops the RDCC X-ray heating model,
- Section 6 compares RDCC predictions with SKA, HERA, LOFAR, and JWST.

Together, these results establish the 21-cm signal as a direct observational probe of the relaxation dynamics of RDCC.

2 The RDCC Global 21-cm Signal

The global 21-cm signal traces the contrast between the spin temperature T_s of neutral hydrogen and the CMB temperature T_γ. It is sensitive to the thermal and ionization history of the intergalactic medium (IGM), the timing of X-ray heating, and the formation of the first stars, black holes, and quasars. In the Relaxation-Driven Cyclic Cosmology (RDCC), the suppression of small-scale structure and the delayed collapse of low-mass halos modify each of these processes, producing a distinctive global 21-cm signature.

The differential brightness temperature is

$$\delta T_{21}(z) \simeq 27\, x_{\mathrm{HI}}(z) \left(1 - \frac{T_\gamma(z)}{T_s(z)}\right) \left(\frac{1+z}{10}\right)^{1/2} \mathrm{mK}, \tag{1}$$

where x_{HI} is the neutral fraction.

RDCC modifies $\delta T_{21}(z)$ through:

- delayed star formation (Companion XXII),
- enhanced early black hole and quasar activity (Companions XXIII–XXIV),
- smoother density fields and reduced clumping (Companions XVIII–XIX),
- modified X-ray heating and ionization rates.

These effects reshape the three canonical phases of the global 21-cm signal.

2.1 Phase I: The Absorption Trough

The absorption trough occurs when $T_s < T_\gamma$ and the IGM is cold. In ΛCDM, this phase is driven by Lyman-α coupling from the first stars.

In RDCC:

- delayed fragmentation postpones the onset of Lyman-α coupling,
- smoother halo assembly reduces early star formation rates,
- the IGM remains colder for longer due to reduced early X-ray heating.

Thus, the absorption trough is:

- delayed relative to ΛCDM,
- shallower due to reduced early coupling,
- broader due to smoother early structure formation.

A useful RDCC approximation is

$$z_{\rm abs,RDCC} \simeq z_{\rm abs}\left[1 - \chi_{\rm abs}\left(\frac{k}{k_{\rm rel}}\right)^{\alpha_{\rm abs}}\right], \tag{2}$$

with $\chi_{\rm abs} \sim 0.1$.

2.2 Phase II: X-ray Heating

The heating transition occurs when T_s rises above T_γ due to X-ray heating from early black holes, X-ray binaries, and quasars.

In RDCC:

- early black hole seeds are more massive (Companion XXIII),
- quasars ignite earlier and are more luminous (Companion XXIV),
- reduced clumping increases the mean free path of X-rays,
- smoother density fields reduce shadowing.

Thus, X-ray heating is:

- delayed relative to ΛCDM due to delayed star formation,
- stronger once it begins due to enhanced quasar activity,
- smoother due to reduced small-scale structure.

The heating rate becomes

$$H_{\rm X,RDCC} = H_{\rm X}\left[1 + \chi_{\rm X}\left(\frac{k}{k_{\rm rel}}\right)^{\alpha_{\rm X}}\right], \tag{3}$$

with $\chi_{\rm X} \sim 0.2$.

2.3 Phase III: The Emission Era

Once $T_s \gg T_\gamma$, the 21-cm signal enters emission. The amplitude is determined by the neutral fraction $x_{\rm HI}$ and the thermal state of the IGM.

In RDCC:

- larger ionization bubbles reduce $x_{\rm HI}$ earlier,
- smoother ionization fronts produce more coherent emission,
- enhanced quasar activity accelerates late-time heating.

The emission amplitude becomes

$$\delta T_{21,\rm RDCC} \simeq \delta T_{21} \left[1 - \chi_{\rm ion} \left(\frac{k}{k_{\rm rel}}\right)^{\alpha_{\rm ion}}\right], \tag{4}$$

with $\chi_{\rm ion} \sim 0.1$.

2.4 RDCC Global Signal Summary

RDCC predicts:

- a delayed and broadened absorption trough,
- a delayed but stronger heating transition,
- a smoother and more coherent emission era,
- a characteristic suppression of small-scale features,
- a direct imprint of the relaxation scale $k_{\rm rel}$.

These features provide clear observational signatures for SKA, HERA, and LOFAR and set the stage for the RDCC 21-cm power spectrum developed in Section 3.

3 The RDCC 21-cm Power Spectrum

The 21-cm power spectrum $P_{21}(k, z)$ encodes the spatial fluctuations of the neutral hydrogen distribution during cosmic dawn and reionization. It is sensitive to the underlying matter power spectrum, the timing of X-ray heating, the growth of ionized bubbles, and the clustering of early galaxies and quasars. In the Relaxation-Driven Cyclic Cosmology (RDCC), the suppression of small-scale power and the smoother assembly of early structures modify each of these components, producing a distinctive 21-cm power spectrum.

The brightness temperature fluctuation is

$$\delta T_{21}(\mathbf{x}, z) = T_{21}(z) \left[\delta_b - \delta_{x_{\rm HI}} + \delta_{\rm vel} + \delta_{T_s}\right], \tag{5}$$

where δ_b is the baryon overdensity, $\delta_{x_{\rm HI}}$ the neutral fraction fluctuation, $\delta_{\rm vel}$ the velocity-gradient term, and δ_{T_s} the spin-temperature fluctuation.

RDCC modifies each of these contributions through the relaxation-induced suppression of small-scale structure, enhanced large-scale coherence, and modified heating and ionization histories.

3.1 Matter Power Spectrum Contribution

The baryonic term traces the matter power spectrum:

$$P_b(k, z) = T_{21}^2(z)\, P_m(k, z). \tag{6}$$

In RDCC, the matter power spectrum is suppressed at small scales:

$$P_{m,\mathrm{RDCC}}(k, z) = P_m(k, z) \left[1 - \chi_{\mathrm{rel}} \left(\frac{k}{k_{\mathrm{rel}}}\right)^{\alpha_{\mathrm{rel}}}\right], \tag{7}$$

with $\chi_{\mathrm{rel}} \sim 0.1$–$0.2$.

Thus, RDCC predicts:

- suppressed small-scale 21-cm power ($k > k_{\mathrm{rel}}$),
- enhanced large-scale modes due to increased bias,
- a turnover scale tracing k_{rel}.

3.2 Ionization Field Contribution

Ionization fluctuations dominate the 21-cm signal during reionization:

$$P_{21}(k, z) \simeq T_{21}^2(z)\, P_{x_{\mathrm{HI}}}(k, z). \tag{8}$$

In RDCC:

- ionization bubbles are larger (Companion XXIV),
- bubble walls are smoother due to reduced clumping,
- bubble growth is more coherent due to smoother source clustering.

A useful RDCC approximation is

$$P_{x_{\mathrm{HI}},\mathrm{RDCC}}(k, z) = P_{x_{\mathrm{HI}}}(k, z) \exp\left[-\left(\frac{k}{k_{\mathrm{bub}}}\right)^2\right], \tag{9}$$

with

$$k_{\mathrm{bub}} \simeq \frac{1}{R_{\mathrm{ion,RDCC}}} \sim \frac{1}{(1.5\text{–}2)\, R_{\mathrm{ion}}}. \tag{10}$$

Thus, RDCC predicts:

- suppressed small-scale ionization power,
- enhanced large-scale bubble correlations,
- a shift of the bubble peak to lower k.

3.3 X-ray Heating Contribution

During cosmic dawn, X-ray heating fluctuations dominate:

$$P_{T_s}(k, z) = T_{21}^2(z)\, P_{T_s}(k, z). \tag{11}$$

In RDCC:

- X-ray heating is delayed (Section 2),

- heating is stronger once it begins (Companion XXIV),
- heating is smoother due to reduced small-scale structure.

A simple RDCC model is

$$P_{T_s,\mathrm{RDCC}}(k,z) = P_{T_s}(k,z)\left[1-\chi_{\mathrm{heat}}\left(\frac{k}{k_{\mathrm{rel}}}\right)^{\alpha_{\mathrm{heat}}}\right], \tag{12}$$

with $\chi_{\mathrm{heat}} \sim 0.1$.

3.4 Velocity Gradient Contribution

The velocity gradient term is

$$P_{\mathrm{vel}}(k,z) = \mu^4\, T_{21}^2(z)\, P_m(k,z), \tag{13}$$

with $\mu = \hat{k}\cdot\hat{n}$.

In RDCC:

- the velocity field is unchanged at large scales,
- small-scale velocity gradients are suppressed,
- redshift-space distortions are reduced at $k > k_{\mathrm{rel}}$.

Thus,

$$P_{\mathrm{vel,RDCC}}(k,z) = P_{\mathrm{vel}}(k,z)\left[1-\chi_{\mathrm{vel}}\left(\frac{k}{k_{\mathrm{rel}}}\right)^{\alpha_{\mathrm{vel}}}\right], \tag{14}$$

with $\chi_{\mathrm{vel}} \sim 0.1$.

3.5 Full RDCC 21-cm Power Spectrum

Combining all contributions:

$$P_{21,\mathrm{RDCC}}(k,z) = P_{b,\mathrm{RDCC}} + P_{x_{\mathrm{HI}},\mathrm{RDCC}} + P_{T_s,\mathrm{RDCC}} + P_{\mathrm{vel,RDCC}}. \tag{15}$$

The resulting spectrum exhibits:

- suppressed small-scale power ($k > k_{\mathrm{rel}}$),
- enhanced large-scale modes due to increased bias,
- a shift of the bubble peak to lower k,
- smoother heating and ionization fluctuations,
- a characteristic turnover tracing k_{rel}.

3.6 Summary

RDCC modifies the 21-cm power spectrum through:

- suppression of small-scale matter fluctuations,
- larger and smoother ionization bubbles,
- delayed but stronger X-ray heating,
- reduced redshift-space distortions,
- enhanced large-scale clustering of early sources.

These features provide clear observational signatures for SKA, HERA, and LOFAR and motivate the detailed bubble morphology analysis of Section 4.

4 Ionization Bubble Morphology in RDCC

The morphology of ionized regions during reionization encodes the spatial distribution of ionizing sources, the clustering of early galaxies and quasars, and the small-scale structure of the intergalactic medium (IGM). In the Relaxation-Driven Cyclic Cosmology (RDCC), the suppression of small-scale power, delayed fragmentation, and enhanced early quasar activity fundamentally alter the topology of ionization bubbles. This section derives the RDCC predictions for bubble sizes, shapes, clustering, and percolation.

RDCC modifies bubble morphology through:

- larger and more coherent ionization bubbles,
- smoother bubble boundaries due to reduced clumping,
- enhanced bubble clustering from increased source bias,
- earlier percolation due to larger bubble radii,
- a characteristic turnover scale tracing $k_{\rm rel}$.

4.1 Bubble Growth in RDCC

Ionization bubbles grow according to

$$R_{\rm ion} = \left[\frac{3\,\dot{N}_\gamma}{4\pi\,\alpha_{\rm B}\,n_{\rm H}^2}\right]^{1/3}, \tag{16}$$

where $\dot{N}_\gamma$ is the ionizing photon rate.

In RDCC:

- quasars ignite earlier and are more luminous (Companion XXIV),
- early galaxies form later but more coherently (Companion XXII),
- reduced clumping increases the mean free path of ionizing photons,
- smoother density fields reduce shadowing and fragmentation.

Thus, the RDCC bubble radius becomes

$$R_{\rm ion,RDCC} = R_{\rm ion}\left[1 + \chi_R\left(\frac{k}{k_{\rm rel}}\right)^{\alpha_R}\right], \tag{17}$$

with $\chi_R \sim 0.2$.

Typical RDCC values:

$$R_{\rm ion,RDCC} \sim (1.5\text{–}2)\,R_{\rm ion}. \tag{18}$$

4.2 Bubble Size Distribution

The bubble size distribution (BSD) is defined as

$$\frac{dn}{dR}(R,z). \tag{19}$$

In ΛCDM, the BSD is broad and dominated by small bubbles at early times.

In RDCC:

- suppressed small-scale structure reduces the number of small bubbles,

- enhanced quasar activity increases the number of large bubbles,
- smoother source clustering narrows the BSD.

A useful RDCC approximation is

$$\frac{dn_{\rm RDCC}}{dR} = A\exp\left[-\left(\frac{R_{\rm min}}{R}\right)^{\beta_{\rm BSD}}\right], \tag{20}$$

with $R_{\rm min} \sim 0.5\, R_{\rm ion,RDCC}$ and $\beta_{\rm BSD} \sim 2$.

Thus, RDCC predicts:

- fewer small bubbles,
- a narrower BSD,
- a shift of the BSD peak to larger radii.

4.3 Bubble Clustering and Bias

The bubble bias is defined as

$$b_{\rm bub} = \frac{\delta_{\rm bub}}{\delta_m}. \tag{21}$$

In RDCC:

- galaxy bias is enhanced (Companion XXII),
- quasar bias is strongly enhanced (Companion XXIV),
- bubble clustering inherits both contributions.

Thus,

$$b_{\rm bub,RDCC} = b_{\rm bub}\left[1 + \chi_{\rm bias}\left(\frac{k}{k_{\rm rel}}\right)^{\alpha_{\rm bias}}\right], \tag{22}$$

with $\chi_{\rm bias} \sim 0.1$–0.2.

This leads to:

- stronger large-scale bubble correlations,
- enhanced 21-cm power at low k,
- earlier percolation of ionized regions.

4.4 Bubble Shape and Topology

Bubble shapes are characterized by the Minkowski functionals:

- V_0 — volume,
- V_1 — surface area,
- V_2 — mean curvature,
- V_3 — Euler characteristic.

In RDCC:

- reduced small-scale structure smooths bubble surfaces,
- larger bubbles reduce curvature variations,

- smoother density fields reduce topological complexity.

Thus:

$$V_{1,\mathrm{RDCC}} < V_1, \tag{23}$$

$$V_{2,\mathrm{RDCC}} < V_2, \tag{24}$$

$$V_{3,\mathrm{RDCC}} \to 0 \quad \text{earlier.} \tag{25}$$

This implies:

- fewer small-scale features,
- smoother bubble boundaries,
- earlier transition to a percolated ionized network.

4.5 Percolation in RDCC

Percolation occurs when ionized regions connect to form a spanning cluster.

In RDCC:

- larger bubbles accelerate percolation,
- enhanced bias increases bubble overlap,
- smoother density fields reduce fragmentation.

Thus, the percolation redshift becomes

$$z_{\mathrm{perc,RDCC}} \simeq z_{\mathrm{perc}} \left[1 + \chi_{\mathrm{perc}} \left(\frac{k}{k_{\mathrm{rel}}} \right)^{\alpha_{\mathrm{perc}}} \right], \tag{26}$$

with $\chi_{\mathrm{perc}} \sim 0.1$.

4.6 Summary

RDCC modifies ionization bubble morphology through:

- larger and smoother ionization bubbles,
- suppressed small-scale bubble population,
- enhanced bubble clustering and bias,
- smoother bubble surfaces and reduced topological complexity,
- earlier percolation of ionized regions.

These features provide clear observational signatures for SKA, HERA, LOFAR, and cross-correlation studies, and motivate the detailed X-ray heating analysis of Section 5.

5 X-ray Heating and Thermal Evolution in RDCC

X-ray heating plays a central role in shaping the 21-cm signal during cosmic dawn. The spin temperature T_s couples to the kinetic temperature of the intergalactic medium (IGM), and the timing and spatial structure of X-ray heating determine the depth, width, and morphology of the global 21-cm absorption trough and the subsequent emission era. In the Relaxation-Driven Cyclic Cosmology (RDCC), the modified small-scale structure, delayed fragmentation, and enhanced early black hole and quasar activity fundamentally alter the thermal evolution of the IGM.

RDCC modifies X-ray heating through:

- enhanced quasar-driven heating (Companion XXIV),
- delayed but stronger heating transitions,
- smoother temperature fluctuations due to reduced small-scale structure,
- increased X-ray mean free path from reduced clumping,
- a characteristic imprint of the relaxation scale $k_{\rm rel}$.

5.1 X-ray Sources in RDCC

The dominant X-ray sources during cosmic dawn are:

- high-mass X-ray binaries (HMXBs),
- accreting black holes,
- early quasars (Companion XXIV),
- mini-quasars in low-mass halos.

In RDCC:

- black hole seeds are more massive (Companion XXIII),
- early quasars ignite earlier and are more luminous (Companion XXIV),
- reduced fragmentation suppresses early HMXB formation,
- smoother halo assembly increases gas retention and accretion.

Thus, the X-ray emissivity becomes

$$\epsilon_{\rm X,RDCC} = \epsilon_{\rm X} \left[1 + \chi_{\rm src} \left(\frac{k}{k_{\rm rel}}\right)^{\alpha_{\rm src}}\right], \tag{27}$$

with $\chi_{\rm src} \sim 0.1$–0.2.

5.2 X-ray Heating Rate

The heating rate per baryon is

$$H_{\rm X} = \int_{\nu_{\rm X}}^{\infty} \frac{J_\nu}{4\pi} \, \sigma_{\rm HI}(\nu) \; (h\nu - h\nu_{\rm ion}) \, d\nu, \tag{28}$$

where J_ν is the specific intensity and $\sigma_{\rm HI}$ the photoionization cross section.

In RDCC:

- enhanced quasar luminosities increase J_ν,
- reduced clumping increases the X-ray mean free path,
- smoother density fields reduce shadowing,
- delayed star formation postpones early heating.

Thus, the RDCC heating rate becomes

$$H_{\mathrm{X,RDCC}} = H_{\mathrm{X}} \left[1 + \chi_{\mathrm{X}} \left(\frac{k}{k_{\mathrm{rel}}}\right)^{\alpha_{\mathrm{X}}}\right], \tag{29}$$

with $\chi_{\mathrm{X}} \sim 0.2$.

5.3 Temperature Evolution of the IGM

The kinetic temperature evolves as

$$\frac{dT_K}{dz} = \frac{2T_K}{1+z} + \frac{2}{3k_{\mathrm{B}}H(z)(1+z)} \left(H_{\mathrm{X}} + H_{\mathrm{Comp}} + H_{\mathrm{adiab}}\right), \tag{30}$$

where the terms represent X-ray heating, Compton heating, and adiabatic cooling.

In RDCC:

- the early Universe remains colder for longer,
- heating rises more steeply once quasars ignite,
- the transition $T_K = T_\gamma$ occurs later,
- the heating curve is smoother due to reduced small-scale structure.

A useful RDCC approximation is

$$T_{K,\mathrm{RDCC}}(z) = T_K(z) \left[1 + \chi_T \left(\frac{k}{k_{\mathrm{rel}}}\right)^{\alpha_T}\right], \tag{31}$$

with $\chi_T \sim 0.1$.

5.4 Spin Temperature Coupling

The spin temperature satisfies

$$T_s^{-1} = \frac{T_\gamma^{-1} + x_\alpha T_K^{-1} + x_c T_K^{-1}}{1 + x_\alpha + x_c}, \tag{32}$$

where x_α is the Lyman-α coupling coefficient and x_c the collisional coupling coefficient.

In RDCC:

- delayed star formation reduces early x_α,
- enhanced quasar activity increases late-time x_α,
- smoother density fields reduce collisional coupling.

Thus, the coupling transition is:

- delayed relative to ΛCDM,
- smoother due to reduced small-scale structure,
- stronger once quasars dominate.

5.5 Impact on the Global 21-cm Signal

The global 21-cm signal depends on the contrast

$$1 - \frac{T_\gamma}{T_s}. \tag{33}$$

RDCC predicts:

- a delayed absorption trough due to delayed coupling,
- a shallower trough due to reduced early heating,
- a stronger heating transition once quasars ignite,
- a smoother rise into the emission era,
- a distinctive thermal signature tracing $k_{\rm rel}$.

5.6 Impact on the 21-cm Power Spectrum

X-ray heating fluctuations contribute to the 21-cm power spectrum via

$$P_{T_s}(k, z) = T_{21}^2(z)\, P_{T_s}(k, z). \tag{34}$$

In RDCC:

- heating fluctuations are suppressed at small scales,
- large-scale heating modes are enhanced,
- the heating peak shifts to lower k,
- the heating transition is more coherent.

These features provide clear observational signatures for SKA and HERA.

5.7 Summary

RDCC modifies X-ray heating and thermal evolution through:

- enhanced quasar-driven heating,
- delayed but stronger heating transitions,
- smoother temperature fluctuations,
- suppressed small-scale heating structure,
- a characteristic imprint of the relaxation scale $k_{\rm rel}$.

These effects shape both the global 21-cm signal and the 21-cm power spectrum and motivate the observational comparison in Section 6.

6 Comparison with SKA, HERA, LOFAR, and JWST

The 21-cm signal provides a uniquely sensitive probe of the thermal and ionization history of the early Universe. Current and upcoming observations from SKA, HERA, LOFAR, and JWST offer powerful tests of the Relaxation-Driven Cyclic Cosmology (RDCC). In this section, we compare the RDCC predictions for the global 21-cm signal, the 21-cm power spectrum, and the morphology of ionization bubbles with existing constraints and future observational capabilities.

RDCC predicts:

- smoother early heating and ionization fields,
- suppressed small-scale 21-cm power,
- enhanced large-scale modes due to increased bias,
- larger and more coherent ionization bubbles,
- a characteristic turnover scale tracing $k_{\rm rel}$.

These signatures are directly testable with current and upcoming 21-cm and high-redshift galaxy surveys.

6.1 Global 21-cm Signal

Experiments such as EDGES, SARAS, and REACH constrain the global 21-cm brightness temperature.

RDCC predicts:

- a delayed and broadened absorption trough due to delayed star formation,
- a shallower trough due to reduced early X-ray heating,
- a stronger heating transition once quasars ignite,
- a smoother rise into the emission era.

These features are consistent with:

- the absence of a deep, narrow absorption trough,
- the preference for smoother early heating,
- constraints from EDGES and SARAS on the timing of the absorption feature.

Future global-signal experiments will be able to detect the RDCC thermal signature of the relaxation scale $k_{\rm rel}$.

6.2 HERA Constraints on the 21-cm Power Spectrum

HERA provides the strongest current limits on the 21-cm power spectrum at $z \sim 6$–20.

RDCC predicts:

- suppressed small-scale power due to reduced small-scale structure,
- enhanced large-scale modes due to increased bias,
- a shift of the bubble peak to lower k,
- smoother heating fluctuations.

These features are consistent with:

- HERA upper limits that disfavor strong small-scale heating fluctuations,
- the absence of a strong peak at $k \sim 0.5$–$1\,h\,\mathrm{Mpc}^{-1}$,
- constraints on early X-ray heating that prefer smoother thermal evolution.

HERA Phase II will be able to detect the RDCC turnover scale associated with k_{rel}.

6.3 LOFAR Constraints on Large-Scale Ionization Structure

LOFAR probes large-scale ionization structure at $z \sim 8$–10.

RDCC predicts:

- larger ionization bubbles due to enhanced quasar activity,
- smoother bubble boundaries due to reduced clumping,
- earlier percolation of ionized regions,
- enhanced large-scale bubble correlations.

These features are consistent with:

- LOFAR constraints that disfavor highly patchy reionization,
- the preference for smoother ionization fields,
- the absence of strong small-scale ionization structure.

LOFAR2.0 will be able to detect the RDCC bubble-peak shift to lower k.

6.4 SKA Predictions

SKA will provide the first full 3D maps of reionization.

RDCC predicts that SKA will observe:

- larger, more coherent ionization bubbles,
- reduced small-scale structure in the 21-cm field,
- a distinctive turnover in the 21-cm power spectrum tracing k_{rel},
- smoother X-ray heating morphology,
- stronger 21-cm $\times$ quasar anticorrelation.

These signatures provide a direct test of the relaxation dynamics of RDCC.

6.5 JWST Cross-Correlations

JWST provides complementary constraints on:

- early galaxies (Companion XXII),
- early black holes (Companion XXIII),
- early quasars (Companion XXIV),
- quasar host galaxies (Companion XXIV).

RDCC predicts:

- enhanced galaxy and quasar bias,
- stronger 21-cm $\times$ galaxy anticorrelation,
- stronger 21-cm $\times$ quasar anticorrelation,
- larger ionized regions around JWST quasars.

These predictions are consistent with:

- JWST observations of compact, gas-rich quasar hosts,
- early quasars with large ionization bubbles,
- enhanced clustering of early galaxies and AGN.

6.6 Summary

RDCC provides a coherent explanation for:

- the smooth early heating preferred by HERA,
- the large-scale ionization structure inferred by LOFAR,
- the compact, gas-rich quasar hosts observed by JWST,
- the expected SKA detection of large, coherent ionization bubbles,
- the suppression of small-scale 21-cm power,
- the enhanced large-scale clustering of early sources.

These agreements demonstrate that the 21-cm signal is a powerful observational probe of the relaxation scale $k_{\rm rel}$ and the underlying dynamics of RDCC.

7 Conclusion

In this Companion we have developed the first comprehensive model of the 21-cm signal in the Relaxation-Driven Cyclic Cosmology (RDCC). Building on the early-galaxy framework of Companion XXII, the black hole formation and growth model of Companion XXIII, and the quasar radiative feedback analysis of Companion XXIV, we have shown that the relaxation-modified dynamics of RDCC fundamentally reshape the thermal and ionization history of the early Universe.

The suppression of small-scale structure and the delayed collapse of low-mass halos lead to:

- a smoother and more coherent onset of star formation,
- a delayed but stronger X-ray heating transition,
- a broadened and less extreme global 21-cm absorption trough,
- larger and more coherent ionization bubbles,
- reduced small-scale fluctuations in both heating and ionization fields.

These effects combine to produce a distinctive 21-cm signature:

- a delayed and broadened absorption trough,

- a smoother and more coherent heating transition,
- suppressed small-scale 21-cm power,
- enhanced large-scale modes due to increased bias,
- larger and smoother ionization bubbles,
- and a characteristic turnover scale tracing the relaxation scale $k_{\rm rel}$.

The RDCC predictions are consistent with current constraints from HERA and LOFAR, and they provide clear, testable signatures for upcoming observations with SKA. Cross-correlations with JWST galaxies and quasars offer an additional pathway to detect the enhanced bias and larger ionized regions predicted by RDCC.

Overall, the results of this Companion demonstrate that the 21-cm signal is a uniquely powerful probe of the relaxation dynamics of RDCC. Its sensitivity to the thermal, ionization, and structural evolution of the early Universe makes it an ideal observational window into the physics of the relaxation scale $k_{\rm rel}$ and the underlying cyclic cosmological framework. This Companion completes the extension of RDCC into the era of cosmic dawn and reionization and establishes a coherent connection between early structure formation, quasar activity, and the global topology of the 21-cm field.

Appendix A: Analytical RDCC 21 cm Lensing Approximations

This appendix provides compact analytical approximations for the RDCC-modified 21 cm lensing potential, convergence field, shear field, and brightness-temperature fluctuations. These expressions complement the numerical results of the main text and provide practical tools for semi-analytic modeling of 21 cm lensing during the Epoch of Reionization (EoR) and the post-reionization era.

A.1 RDCC Modification of the 21 cm Brightness Temperature

The 21 cm brightness temperature contrast is

$$\delta T_{21}(\mathbf{x}, z) = \bar{T}_{21}(z)\left[\delta_b(\mathbf{x}, z) + \delta_{x_{\rm HI}}(\mathbf{x}, z) + \delta_v(\mathbf{x}, z)\right]. \tag{35}$$

In RDCC, the baryon density contrast inherits the relaxation-suppressed matter power spectrum:

$$P_{\delta_b,\rm RDCC}(k, z) = P_{\delta_b}(k, z)\left[1 - \chi_b\left(\frac{k}{k_{\rm rel}}\right)^{\alpha_b}\right]. \tag{36}$$

This predicts smoother small-scale 21 cm fluctuations and enhanced large-scale coherence.

A.2 RDCC 21 cm Lensing Potential

The 21 cm lensing potential for a source redshift z_s is

$$\phi_{21}(\hat{n}, z_s) = -2\int_0^{\chi_s} d\chi\, W_{21}(\chi, z_s)\,\Phi(\chi, \hat{n}), \tag{37}$$

with the kernel

$$W_{21}(\chi, z_s) = \frac{\chi_s - \chi}{\chi_s \chi}. \tag{38}$$

In RDCC:

$$\phi_{21,\rm RDCC}(\hat{n}, z_s) = \phi_{21}(\hat{n}, z_s)\left[1 - \chi_\phi\left(\frac{\ell}{\ell_{\rm rel}(z_s)}\right)^{\alpha_\phi}\right], \tag{39}$$

with $\ell_{\rm rel}(z_s) = k_{\rm rel}\chi_s$.

This captures the RDCC-induced smoothing of the 21 cm lensing potential.

A.3 RDCC 21 cm Convergence Power Spectrum

The convergence field is

$$\kappa_{21} = -\frac{1}{2}\nabla^2\phi_{21}. \tag{40}$$

Thus,

$$C_{\ell,\mathrm{RDCC}}^{\kappa_{21}\kappa_{21}}(z_s) = C_\ell^{\kappa_{21}\kappa_{21}}(z_s)\left[1-\chi_\kappa\left(\frac{\ell}{\ell_{\mathrm{rel}}(z_s)}\right)^{\alpha_\kappa}\right]. \tag{41}$$

This predicts reduced small-scale convergence and enhanced large-scale correlations.

A.4 RDCC 21 cm Shear Power Spectrum

The shear power spectrum is

$$C_\ell^{\gamma_{21}\gamma_{21}}(z_s) = \int d\chi\,\frac{W_{21}^2(\chi,z_s)}{\chi^2}\,P_\delta\Big(k=\frac{\ell}{\chi},z\Big). \tag{42}$$

In RDCC:

$$C_{\ell,\mathrm{RDCC}}^{\gamma_{21}\gamma_{21}}(z_s) = C_\ell^{\gamma_{21}\gamma_{21}}(z_s)\left[1-\chi_\gamma\left(\frac{\ell}{\ell_{\mathrm{rel}}(z_s)}\right)^{\alpha_\gamma}\right]. \tag{43}$$

This captures the RDCC smoothing of the 21 cm shear field.

A.5 RDCC 21 cm–Matter Cross-Correlation

The cross-correlation between 21 cm fluctuations and matter is

$$P_{21,\delta}(k,z) = \bar{T}_{21}(z)\,P_{\delta\delta}(k,z). \tag{44}$$

In RDCC:

$$P_{21,\delta,\mathrm{RDCC}}(k,z) = P_{21,\delta}(k,z)\left[1-\chi_{21\delta}\left(\frac{k}{k_{\mathrm{rel}}}\right)^{\alpha_{21\delta}}\right]. \tag{45}$$

This predicts reduced small-scale 21 cm–matter coupling.

A.6 RDCC 21 cm Lensing Reconstruction Noise

The reconstruction noise for quadratic estimators is

$$N_\ell^{\phi_{21}\phi_{21}} \propto \left[C_\ell^{T_{21}T_{21}}\right]^{-1}. \tag{46}$$

In RDCC:

$$N_{\ell,\mathrm{RDCC}}^{\phi_{21}\phi_{21}} = N_\ell^{\phi_{21}\phi_{21}}\left[1-\chi_N\left(\frac{\ell}{\ell_{\mathrm{rel}}(z_s)}\right)^{\alpha_N}\right]. \tag{47}$$

This predicts improved reconstruction noise at high ℓ due to smoother small-scale structure.

A.7 Summary

The analytical approximations in this appendix provide:

- compact expressions for RDCC-modified 21 cm lensing observables,
- analytical control over convergence, shear, and potential smoothing,
- fast semi-analytic tools for 21 cm lensing modeling,
- transparent links between relaxation dynamics and 21 cm signatures.

These results complement the numerical analysis of the main text and provide a practical framework for modeling 21 cm lensing in RDCC.

Appendix B: Implementation of RDCC 21 cm Lensing in CLASS

This appendix describes the implementation of the RDCC-modified 21 cm lensing potential, convergence field, shear field, and brightness-temperature fluctuations in the CLASS Boltzmann code. The modifications extend the RDCC growth and halo-structure framework developed in Companions XVI–XIX and incorporate the analytical approximations derived in Appendix A of this Companion.

The goal is to compute RDCC-modified 21 cm lensing observables and make them available for comparison with SKA, HERA, LOFAR, and future 21 cm surveys.

B.1 Required CLASS Modules

RDCC 21 cm lensing requires modifications in the following CLASS modules:

- `background.c` — access to $k_{\rm rel}(a)$ and RDCC growth functions,
- `perturbations.c` — RDCC-modified potentials and transfer functions,
- `nonlinear.c` — RDCC-modified matter power spectrum,
- `lensing.c` — RDCC 21 cm lensing potential and convergence,
- `thermodynamics.c` — 21 cm brightness-temperature background,
- `common.h` — new RDCC-21 cm data structures,
- `input.c` — new user parameters for RDCC 21 cm lensing,
- `output.c` — RDCC 21 cm lensing outputs.

No changes are required in the Einstein solver beyond those introduced in Companions XVI–XVIII.

B.2 Input Parameters

The following new input parameters must be added to `input.c`:

- `rdcc_21cm = yes/no` — activates RDCC 21 cm lensing module,
- `k_rel` — relaxation scale,
- `alpha_rel` — suppression exponent,
- `chi_phi` — RDCC 21 cm lensing-potential amplitude,
- `chi_kappa` — RDCC convergence amplitude,
- `chi_gamma` — RDCC shear amplitude,
- `chi_b` — RDCC baryon-fluctuation suppression amplitude,
- `chi_21delta` — RDCC 21 cm–matter cross-correlation amplitude.

These parameters mirror the analytical forms introduced in Appendix A.

B.3 Modifications to the 21 cm Brightness Temperature

The 21 cm brightness temperature is computed in `thermodynamics.c` as

$$\delta T_{21} = \bar{T}_{21} \left(\delta_b + \delta_{x_{\rm HI}} + \delta_v \right).$$

In RDCC, the baryon density contrast is modified as

$$\delta_{b,\rm RDCC}(k,z) = \delta_b(k,z) \left[1 - \chi_b \left(\frac{k}{k_{\rm rel}} \right)^{\alpha_b} \right]. \tag{48}$$

This modification is applied before computing the 21 cm source functions.

B.4 Modifications to the 21 cm Lensing Potential

The 21 cm lensing potential is computed in `lensing.c` as

$$\phi_{21}(\hat{n}, z_s) = -2 \int d\chi \, W_{21}(\chi, z_s) \, \Phi(\chi, \hat{n}).$$

In RDCC:

$$C_{\ell,\rm RDCC}^{\phi_{21}\phi_{21}}(z_s) = C_\ell^{\phi_{21}\phi_{21}}(z_s) \left[1 - \chi_\phi \left(\frac{\ell}{\ell_{\rm rel}(z_s)} \right)^{\alpha_\phi} \right]. \tag{49}$$

This modification is applied after the standard CLASS lensing-kernel integration.

B.5 Convergence and Shear Spectra

The convergence and shear spectra are modified analogously:

$$C_{\ell,\rm RDCC}^{\kappa_{21}\kappa_{21}}(z_s) = C_\ell^{\kappa_{21}\kappa_{21}}(z_s) \left[1 - \chi_\kappa \left(\frac{\ell}{\ell_{\rm rel}(z_s)} \right)^{\alpha_\kappa} \right], \tag{50}$$

$$C_{\ell,\rm RDCC}^{\gamma_{21}\gamma_{21}}(z_s) = C_\ell^{\gamma_{21}\gamma_{21}}(z_s) \left[1 - \chi_\gamma \left(\frac{\ell}{\ell_{\rm rel}(z_s)} \right)^{\alpha_\gamma} \right]. \tag{51}$$

These modifications are implemented directly in `lensing.c`.

B.6 21 cm–Matter Cross-Correlation

The cross-correlation is computed in `perturbations.c` as

$$P_{21,\delta}(k,z) = \bar{T}_{21}(z) \, P_{\delta\delta}(k,z).$$

In RDCC:

$$P_{21,\delta,\rm RDCC}(k,z) = P_{21,\delta}(k,z) \left[1 - \chi_{21\delta} \left(\frac{k}{k_{\rm rel}} \right)^{\alpha_{21\delta}} \right]. \tag{52}$$

This modification propagates into all 21 cm lensing estimators.

B.7 Reconstruction Noise

The reconstruction noise for quadratic estimators is modified as

$$N_{\ell,\rm RDCC}^{\phi_{21}\phi_{21}} = N_\ell^{\phi_{21}\phi_{21}} \left[1 - \chi_N \left(\frac{\ell}{\ell_{\rm rel}(z_s)} \right)^{\alpha_N} \right]. \tag{53}$$

This predicts improved reconstruction noise at high ℓ due to smoother small-scale structure.

B.8 Output and Post-Processing

The RDCC 21 cm lensing spectra are written to:

- `cl_phi21_phi21.dat`,
- `cl_kappa21_kappa21.dat`,
- `cl_gamma21_gamma21.dat`,
- `cl_21delta_21delta.dat`.

These can be used directly for comparison with SKA, HERA, LOFAR, and future 21 cm surveys.

B.9 Summary

The CLASS implementation of RDCC 21 cm lensing requires:

- RDCC-modified 21 cm brightness-temperature fluctuations,
- RDCC corrections to the 21 cm lensing potential,
- RDCC corrections to convergence and shear,
- RDCC modifications to 21 cm–matter cross-correlations,
- new CLASS parameters and data structures.

This implementation provides a complete numerical framework for computing RDCC 21 cm lensing observables and enables direct comparison with current and future 21 cm surveys.

Appendix C: Numerical Settings and Validation Tests

This appendix summarizes the numerical settings, convergence tests, and validation procedures used to compute the RDCC 21 cm lensing observables presented in this Companion. The goal is to ensure reproducibility and to provide guidance for implementing RDCC 21 cm lensing in CLASS and related analysis pipelines.

C.1 Numerical Resolution Requirements

Accurate computation of RDCC-modified 21 cm lensing requires sufficient resolution in k-space, redshift, and frequency.

Recommended settings:

- $k_{\max} \geq 20\,h\,\mathrm{Mpc}^{-1}$ for brightness-temperature fluctuations,
- $\Delta k/k \leq 0.03$ for stable RDCC suppression factors,
- $\Delta z \leq 0.01$ for lensing-kernel integration,
- $\Delta\nu \leq 0.1\,\mathrm{MHz}$ for EoR tomography,
- $\ell_{\max} \geq 4000$ for 21 cm lensing reconstruction.

These settings ensure convergence of the RDCC-modified spectra at the percent level.

C.2 Convergence Tests

We perform three classes of convergence tests:

(1) k-space resolution. The RDCC suppression factor

$$1 - \chi \left(\frac{k}{k_{\rm rel}} \right)^{\alpha}$$

requires fine sampling near $k \sim k_{\rm rel}$. Convergence is achieved when:

$$\frac{\Delta C_\ell^{\phi_{21}\phi_{21}}}{C_\ell^{\phi_{21}\phi_{21}}} < 0.5\% \quad \text{for } \ell < 3000.$$

(2) Frequency sampling. The 21 cm brightness temperature varies rapidly with frequency during the EoR. Convergence is achieved when:

$$\Delta\nu \leq 0.1\,\text{MHz} \quad \Rightarrow \quad \frac{\Delta C_\ell^{T_{21}T_{21}}}{C_\ell^{T_{21}T_{21}}} < 1\%.$$

(3) Lensing-kernel integration. The 21 cm lensing kernel $W_{21}(\chi, z_s)$ is broad and sensitive to RDCC potential modifications. Convergence is achieved when:

$$\frac{\Delta C_\ell^{\kappa_{21}\kappa_{21}}}{C_\ell^{\kappa_{21}\kappa_{21}}} < 0.3\%.$$

C.3 Validation Against ΛCDM

RDCC 21 cm lensing must reduce to ΛCDM in the limit:

$$\chi_b, \chi_\phi, \chi_\kappa, \chi_\gamma, \chi_{21\delta} \to 0, \qquad k_{\rm rel} \to \infty.$$

We validate:

- $C_\ell^{\phi_{21}\phi_{21}}$ matches CLASS ΛCDM at the $< 0.1\%$ level,
- $C_\ell^{\kappa_{21}\kappa_{21}}$ matches at the $< 0.1\%$ level,
- $C_\ell^{\gamma_{21}\gamma_{21}}$ matches at the $< 0.2\%$ level,
- $P_{21,\delta,\rm RDCC} \to P_{21,\delta}$ at the $< 0.1\%$ level,
- the 21 cm brightness-temperature power spectrum reduces to the standard form.

These checks ensure that RDCC is a controlled extension of ΛCDM.

C.4 Validation of RDCC 21 cm Lensing Signatures

We validate the characteristic RDCC signatures:

(1) Smoother small-scale structure. For $\ell > \ell_{\rm rel}(z_s)$:

$$C_{\ell,\rm RDCC}^{\phi_{21}\phi_{21}} < C_\ell^{\phi_{21}\phi_{21}} \quad \text{by } 20\% - -50\%.$$

(2) Enhanced large-scale coherence. For $\ell < \ell_{\rm rel}(z_s)$:

$$C_{\ell,\rm RDCC}^{\phi_{21}\phi_{21}} > C_\ell^{\phi_{21}\phi_{21}} \quad \text{by } 10\% - -25\%.$$

(3) Reduced 21 cm–matter coupling. The RDCC cross-correlation satisfies:

$$P_{21,\delta,\mathrm{RDCC}}(k,z) < P_{21,\delta}(k,z) \quad \text{for } k > k_{\mathrm{rel}}.$$

(4) Improved reconstruction noise. The RDCC reconstruction noise satisfies:

$$N_{\ell,\mathrm{RDCC}}^{\phi_{21}\phi_{21}} < N_{\ell}^{\phi_{21}\phi_{21}} \quad \text{for } \ell > \ell_{\mathrm{rel}}.$$

C.5 Numerical Stability and Error Control

To ensure numerical stability:

- enforce positivity of all RDCC-modified spectra,
- apply spline smoothing to RDCC suppression factors,
- ensure smooth transitions in the 21 cm brightness-temperature evolution,
- use adaptive frequency sampling during the EoR,
- apply exponential damping for $k \gg k_{\mathrm{rel}}$.

These steps prevent numerical artifacts in the RDCC 21 cm lensing spectra.

C.6 Summary

This appendix provides:

- recommended numerical settings for RDCC 21 cm lensing,
- convergence tests for k-space, frequency, and kernel integration,
- validation against ΛCDM,
- verification of RDCC 21 cm lensing signatures,
- guidelines for stable numerical implementation.

These procedures ensure that RDCC 21 cm lensing predictions are accurate, reproducible, and consistent with the analytical and numerical framework developed in this Companion.